

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 12: Directional & Lateral Stability
(Dutch Roll & Spiral)

Syllabus Source: CAE Oxford (pp. 425-464) | Pass Target: >= 90% AviationExam

1. DIRECTIONAL (WEATHERCOCK) STABILITY

Directional stability is stability about the normal (vertical) axis following a sideslip angle (Beta):

- Weathercock Principle: When an aircraft sideslips, relative wind strikes the fuselage and vertical tail from the side. The vertical stabilizer (fin) generates a horizontal aerodynamic restoring force behind the CG, yawing the nose back into the relative wind.
- Contributions to Directional Stability: Vertical Fin (major positive stabilizing contributor); Rear Fuselage (stabilizing); Swept Wings (stabilizing: retreating wing has higher effective span and profile drag, pulling nose straight); Forward Fuselage and Nacelles (DESTABILIZING: generate adverse destabilizing yaw moments ahead of CG).

2. LATERAL (DIHEDRAL) STABILITY & CROSS-COUPLING

Lateral stability is stability about the longitudinal axis (roll) when perturbed into a sideslip:

- Geometric Dihedral (Gamma): Upward angle of wings from root to tip. When wings roll into a sideslip, the low wing meets relative airflow at a HIGHER effective angle of attack than the high wing, generating more lift and rolling the wings back level!
- Wing Sweepback Effect: Massive contributor to lateral stability! When a swept-wing aircraft sideslips, the wing into the wind (leading wing) has reduced effective sweep ($\Lambda - \beta$), increasing its normal velocity component ($V \cos(\Lambda - \beta)$) and lift, while the trailing wing loses lift. This produces a powerful restoring roll moment into the slip!
- High Wing (Anhedral vs Dihedral): High-mounted wings experience airflow blockage by the fuselage during sideslip, increasing high-wing lift (keel effect). Because swept high-wing air transports (C-17, Antonov) possess EXCESSIVE lateral stability, they use ANHEDRAL (negative dihedral / drooped wings) to restore balance!

3. DYNAMIC MODES: DUTCH ROLL VS SPIRAL DIVERGENCE

Dynamic Mode	Relative Stability Balance	Coupled Flight Motion & Operational Hazard
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Dutch Roll	Excessive Lateral Stability (Dihedral/Sweep) combined with WEAK Directional Stability.	A coupled out-of-phase yaw and roll motion tracing a figure-8 corkscrew trajectory. Annoying to passengers; potentially divergent at high altitudes. MANDATORY REMEDY: Yaw Damper!

Spiral Divergence	STRONG Directional Stability combined with WEAK Lateral Stability (Dihedral).	Aircraft sideslips into turn; strong fin weathercocks nose into turn, while weak dihedral fails to level wings. Bank angle and rate of descent steepen progressively into a high-speed spiral dive. Non-oscillatory; slow divergence; easily corrected by pilot.
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[!] EXAM TRAP (AVIATIONEXAM) - Dutch Roll vs Spiral Stability Trade-off

- Designing for high lateral stability suppresses spiral dive, BUT CREATES DUTCH ROLL! - Designing for high directional stability suppresses Dutch roll, BUT CREATES SPIRAL DIVERGENCE! - Modern transport jets accept mild spiral divergence (easily controlled by autopilot) and eliminate Dutch roll using automatic YAW DAMPERS.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Dutch Roll vs Spiral Divergence

Rule 1 (Dutch Roll): Lateral Stability > Directional Stability -> Dihedral/sweep rights wings too aggressively compared to weak rudder/fin damping, causing coupled out-of-phase yawing and rolling. Countered with Yaw Damper. Rule 2 (Spiral Divergence): Directional Stability > Lateral Stability -> Strong vertical fin aligns nose with wind before dihedral levels wings. Bank angle tightens into a descending spiral. Exam Rule: Dutch Roll = Strong Lateral; Spiral = Strong Directional.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What stability combination causes an aircraft to be susceptible to DUTCH ROLL?

- [A]: Strong directional stability and weak lateral stability
- [B]: Strong LATERAL stability (dihedral effect) combined with relatively WEAK DIRECTIONAL stability
- [C]: Forward CG
- [D]: Zero sweepback

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Dutch roll occurs when roll restoring forces (dihedral/sweep) are much stronger than directional yaw restoring forces. The aircraft over-corrects in roll before yaw can damp out, resulting in a corkscrew oscillation.

Q2: Why do many military cargo aircraft with swept, high-mounted wings incorporate ANHEDRAL (drooped wingtips)?

- [A]: To increase cruise speed
- [B]: To reduce excessive lateral stability created by the combination of high wing position and wing sweep, which would otherwise induce violent Dutch roll
- [C]: To assist engine maintenance
- [D]: To increase take-off lift

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Both swept wings and high-wing placement generate strong lateral stability (effective dihedral). Left unchecked, this makes the aircraft violently prone to Dutch roll. Anhedral is added to reduce roll-due-to-sideslip to a safe balance.

Q3: What is the primary operational function of an aircraft Yaw Damper system?

- [A]: To steer the nosewheel on landing
- [B]: To detect and immediately suppress Dutch Roll oscillations by commanding automatic rudder deflections
- [C]: To prevent stalls
- [D]: To trim the ailerons

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A yaw damper uses rate gyros to sense yaw rate and commands small, fast opposing rudder deflections via servo actuators, damping out Dutch roll without pilot pedal movement.